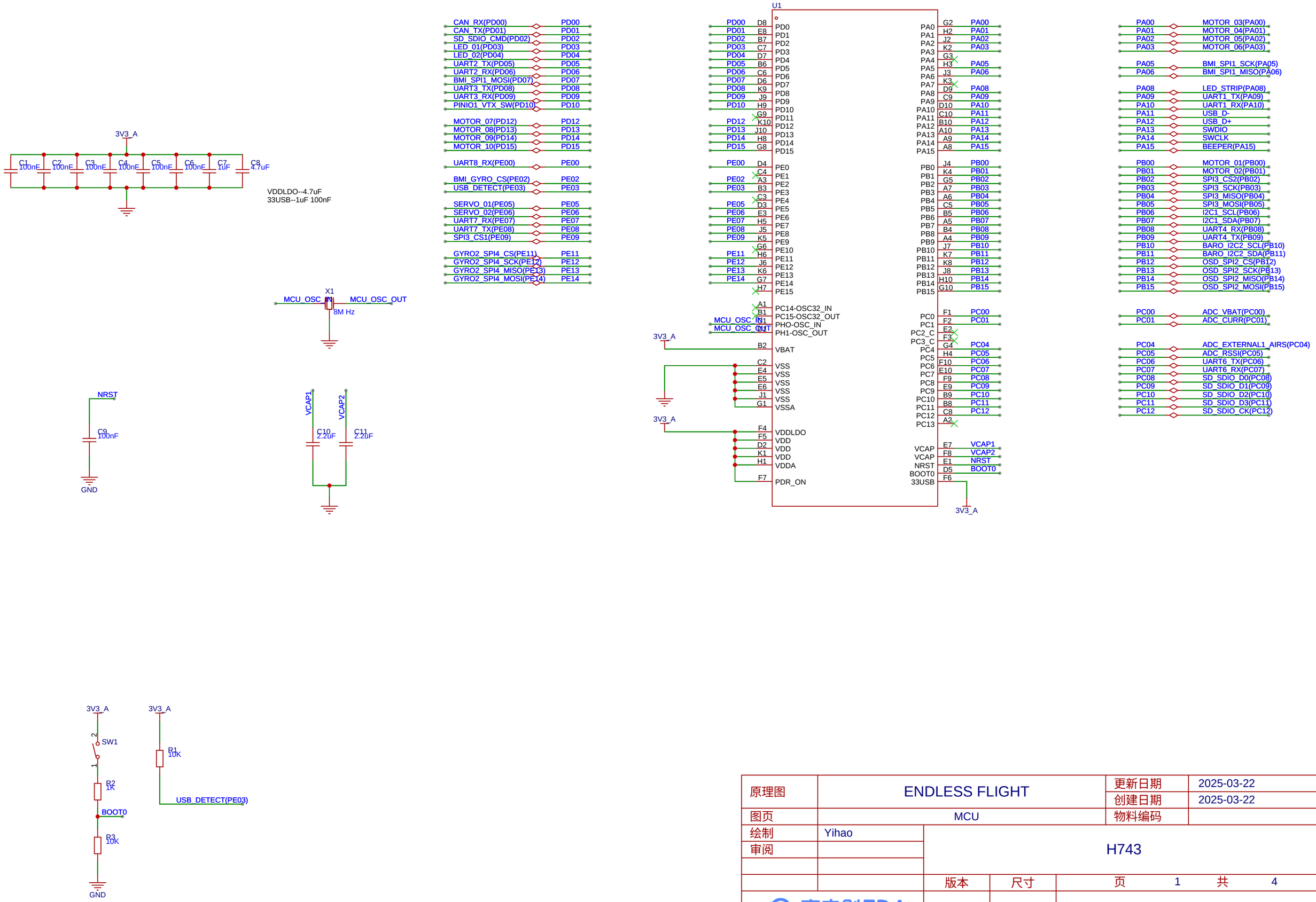
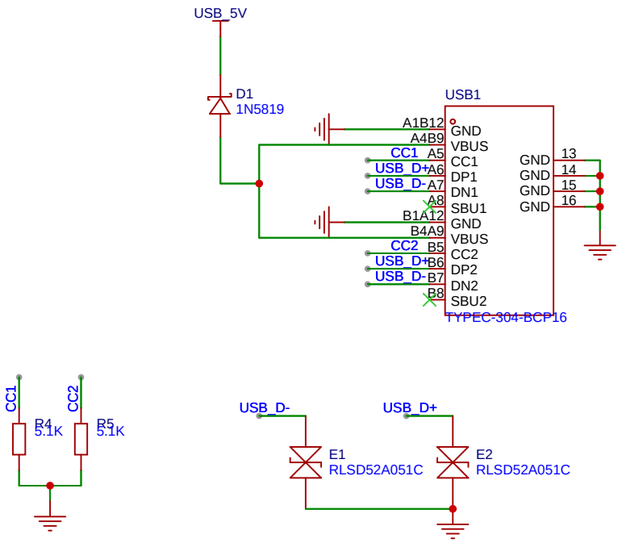
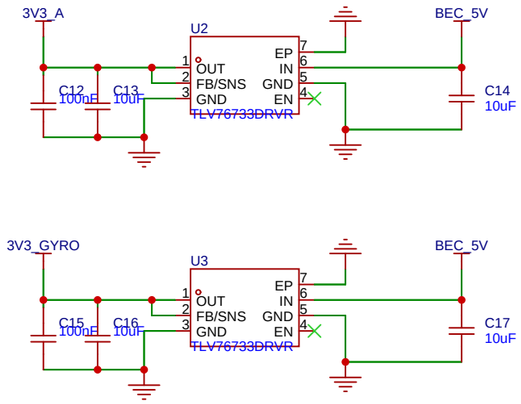


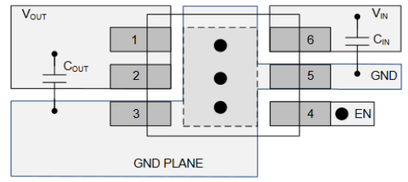
原理图	ENDLESS FLIGHT	更新日期	2025-03-22
		创建日期	2025-03-22
图页	MCU	物料编码	
绘制	Yihao	H743	
审阅			
		版本	尺寸
		页 1 共 4	
嘉立创EDA		MK3.7.2	A2
		KunPeng	

USB OTG



LDO(5V-3.3V)

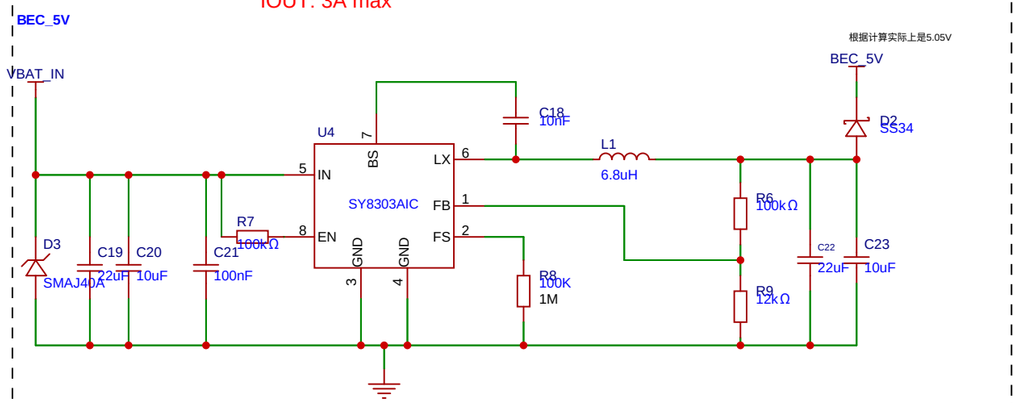


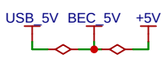


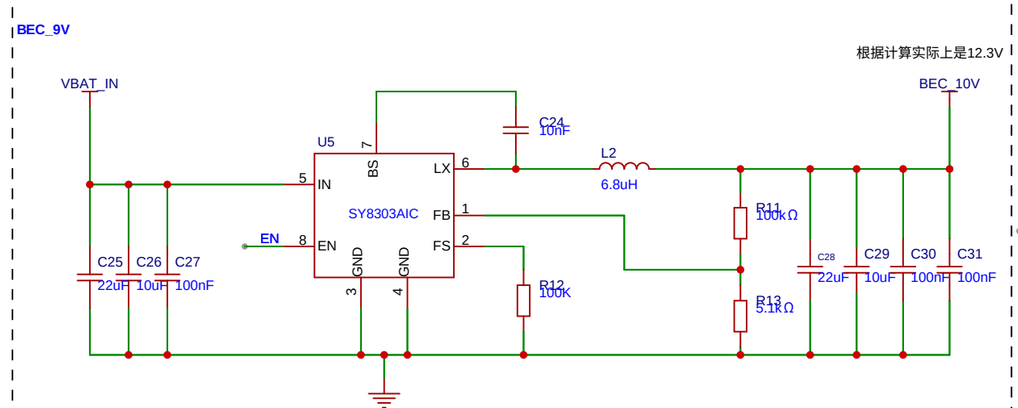
● Represents via used for application-specific connections

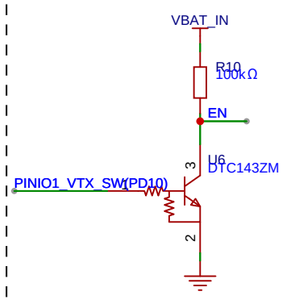
BUCK(同步整流)

VIN: 6S max
VOUT: 5V or 9V
IOUT: 3A max

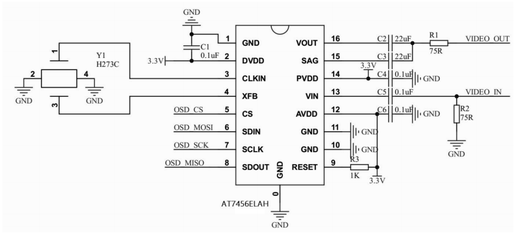


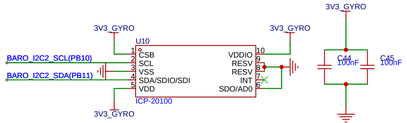






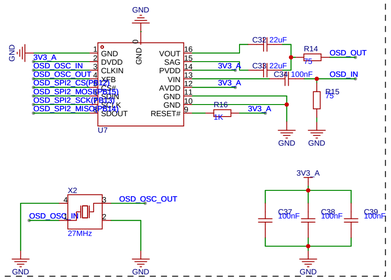
原理图	ENDLESS FLIGHT		更新日期	2025-03-22
			创建日期	2025-03-22
图页	POWER		物料编码	
绘制	Yihao	H743		
审阅				
		版本	尺寸	页 2 共 4
嘉立创EDA		V1.0	A2	SWU



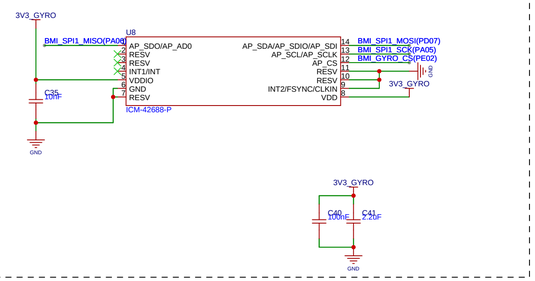


AD0=0 0X63
AD0=1 0X64

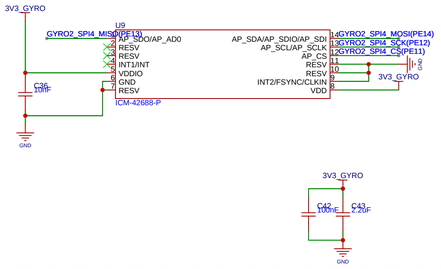
图传(SPI2)



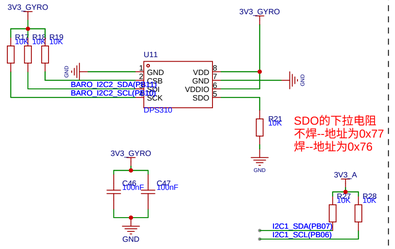
ICM42688P(SPI1)



ICM42688P(SPI4)

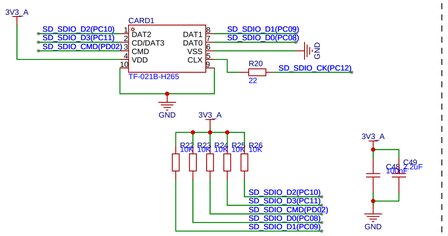


气压计DPS310(I2C2)

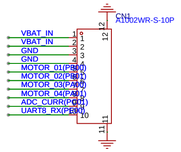


SDO的下拉电阻
不焊--地址为0x77
焊--地址为0x76

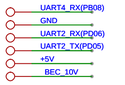
TF(SWDIO)



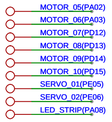
四合一电调接口/4路PWM/电流ADC/RX1



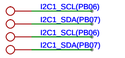
DJI天空端直插接口



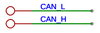
9路PWM/6路电机/2路舵机/1路灯带



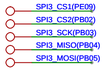
2路I2C



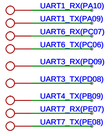
1路CAN



1路SPI3



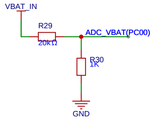
6路串口



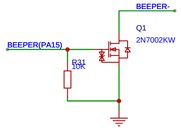
蜂鸣器

2路ADC

电池电压采集

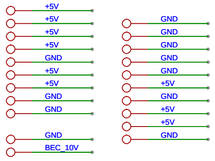


有源蜂鸣器

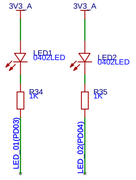


摄像头接口

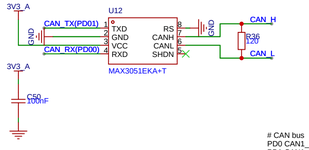
各种电源/5V/10V/VBAT



指示灯



CAN



CAN bus
PD0 CAN1_RX CAN1
PD1 CAN1_TX CAN1
使用GANZ开关，-RX与-TX不用反接

原理图	ENDLESS FLIGHT	更新日期	2025-03-22
		创建日期	2025-03-22
图页	HARDWARE	物料编码	
绘制	Yihao	H743	
审阅			
	版本	尺寸	页 3 共 4
嘉立创EDA	V1.0	A2	SWU

MK4.0 ：
1、优化散热，主要在板子顶层USB下方添加开窗；
2、优化BEC部分，改10VBEC为12VBEC,并把12VBEC的输出线路加宽。将BUCK的开关频率由2.1M降低到1M，减小BEC部分的发热。更换额定电流更大、内阻更低的电感。
3、改FPC底座，由前插翻盖式，改为前插后掀翻盖式；
4、减小30.5M4安装孔位的外径；
5、改小过孔孔径，改最小过孔内径为0.2mm，外径为0.35mm;
6、改512MB的SDNAND为短体高2.5mmTF卡插座；
7、增加USB插入检测PE3：GPIO_OTGFS_VBUS；
8、增加飞控外框切割标识，并开窗处理；
9、改LED电阻为1K，并更改LED的驱动方式；
10、字符增大；
11、给增加AUX接口增加一个CS片选PC14，总共三个，这样可以多接IMU；
12、更换CAN收发芯片型号为MAX3051EKA+T；
13、更换给IMU和气压计供电的LDO型号为TLV76733DRVR

MK4.1 ：
1、优化BEC部分，增大5VBEC的输入线路宽度；
2、优化字体丝印，增大变细些；
3、飞控进DFU不灵敏，BOOT0加个分压电阻试试；
4、第五层铺地处理，电感下方铺地处理；
5、8M晶振向主控方向移动，尽量远离IMU；
6、去掉VB2、CU2的ADC;
7、增添SPI3_CS2(PE09)；
8、去掉NRST引脚的强上拉电阻；

MK5.0 ：
1、改为双IMU ICM42688方案；
2、给两个IMU加屏蔽框；
3、电调回传串口选择RX8

MK5.beta ：
1、将驱动BB响三极管 SS8050改为MOS 2N7002KW；
2、增加一个气压计ICP-20100，使飞控系统为双冗余IMU&Baro；
3、改大焊盘，去掉RX8 TX8 VBAT GND 焊盘；
4、改小过孔孔径，改最小过孔内径为6mil，外径为12mil;
5、给USB加ESD保护；
6、将两个晶振尽量做到包地处理，晶振正下方不放过孔地，四周放